

Exploring Global Health Disparities:
A Statistical Investigation into the Factors Influencing Life Expectancy

Pinar Targil (pt38), Meaghan Ramlakhan (mcr8), Charis Xiong (cx25)

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STAT 410 - Linear Regression

Final Project Report

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Statement of the Problem

The purpose of this project is to investigate the key determinants of life expectancy across countries using the World Health Organization (WHO) Life Expectancy dataset, supplemented with economic indicators from the United Nations. The study aims to build a predictive regression model that accurately forecasts life expectancy by identifying and quantifying the influence of various health, economic, and social factors. With a complex set of potential predictors, this analysis seeks to distill the essential variables while addressing statistical challenges like multicollinearity and overfitting.

This project sets out to solve the following central problems:

- **What factors most strongly influence life expectancy worldwide?**
- **How does development status (developed vs. developing countries) affect average life expectancy?**
- **Can an effective, reliable statistical model be developed to predict life expectancy across different countries, based on available health and economic data?**

The project will explore how health indicators (such as disease prevalence and immunization rates), mortality rates, economic conditions (like GDP), and education levels contribute to differences in life expectancy across 193 countries from 2000 to 2015.

The main goals of this study are:

- To identify and quantify the most influential factors impacting life expectancy across countries.
- To develop a robust and interpretable regression model that can accurately predict life expectancy.
- To perform rigorous statistical validation and diagnostic checks to ensure model reliability.
- To generate actionable insights that can guide public health policy and resource allocation to improve global health outcomes and reduce disparities in longevity.

The impact of development status, mortality rates, and healthcare investment on life expectancy is of particular interest, as well as how various social, economic, and health-related factors interact to create disparities between countries. Gaining insights into these relationships can guide targeted policy decisions aimed at narrowing longevity gaps and enhancing global health outcomes.

Introduction

This study aimed to explore and quantify the primary factors influencing life expectancy worldwide by developing a predictive regression model based on data from the World Health Organization (WHO) and the United Nations. This dataset covered 193 countries over the years 2000 to 2015, containing a broad spectrum of health, economic, and social variables, which include mortality rates, disease prevalence, immunization coverage, healthcare expenditure, GDP, and educational attainment. By investigating these factors, the project aimed to uncover actionable insights into the disparities in life expectancy across countries.

To conceptualize this project, we conducted extensive background research, reviewing peer-reviewed articles from *The Lancet Public Health*, *BMC Public Health*, and the *International Journal of Epidemiology*. These readings highlighted the multifaceted influences on life expectancy, such as healthcare infrastructure, income inequality, disease burden, and education. Our analytical approach was guided by *Applied Linear Regression Models* by Kutner, Nachtsheim, and Neter, which informed our model-building process, diagnostic techniques, and strategies for addressing multicollinearity. We also referenced technical documentation from the WHO and the World Bank to accurately interpret global health indicators and contextualize our findings within established public health frameworks.

We applied multiple linear regression using exhaustive best-subset selection to identify the optimal combination of predictors, tested alternative model structures such as polynomial terms, and validated model assumptions using diagnostic tools including residual plots and statistical tests like the Breusch-Pagan and Shapiro-Wilk tests. Model performance was evaluated using metrics such as adjusted R^2 , AIC, BIC, and prediction intervals to determine the most robust and interpretable model for predicting life expectancy.

Data Collection

The dataset used in this analysis was sourced from Kaggle and originally compiled by the World Health Organization (WHO) and the United Nations. It includes 2,938 observations from 193 countries spanning the years 2000 to 2015, with 22 variables representing health, economic, and social indicators. These include mortality rates (Adult.Mortality, infant.deaths), disease prevalence (HIV/AIDS, Measles), immunization coverage (Hepatitis.B, Polio, Diphtheria), healthcare spending (Total.expenditure, percentage.expenditure), GDP per capita, and average years of schooling.

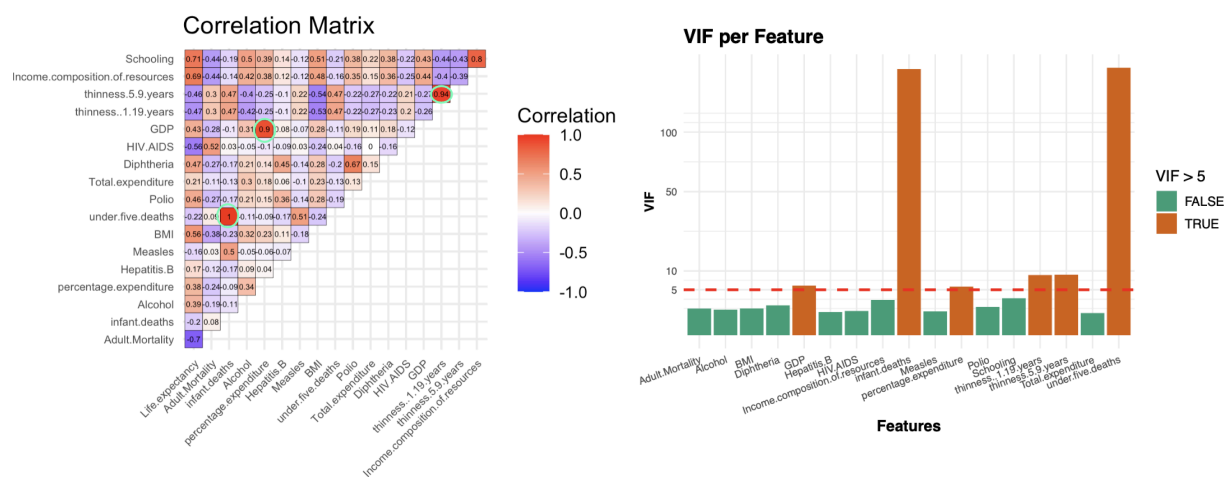
Summary statistics from the dataset revealed that life expectancy across countries ranged from 36.3 to 89 years, with a mean of approximately 69.2 years. Adult mortality had a mean of 164.8 deaths per 1000 population, while infant deaths averaged 30.3 per 1000. Schooling ranged from 0 to 20.7 years, with a mean of 11.99. GDP per capita showed extreme skew, with values ranging from \$1.68 to over \$119,000 and a mean of \$7,483. Measles cases per 1000 population were also highly variable, with a mean of 2,419 and a maximum exceeding 212,000, suggesting strong disparities in healthcare infrastructure. These summary statistics reinforced the need for transformation and variable selection strategies to address skew and heterogeneity in the data.

Our initial plan focused on thorough data exploration and cleaning, including handling missing data, dealing with outliers, and performing exploratory data analysis (EDA) to understand distributions and relationships between variables. During this process, we found that *Population* was not a reliable variable to include in our model for a few reasons: it was highly skewed, had a lot of missing values, and there were noticeable discrepancies across years for the same country that didn't make sense. Because of this, we decided to drop *Population* from our analysis. For the other variables with missing values, we used imputation techniques based on the type of outliers present. Specifically, if a variable had high outliers, we replaced missing values with the median. If a variable had low outliers, we replaced missing values with the mean. This approach helped us preserve the overall structure of the data while handling missingness in a reasonable manner.

We began our analysis by checking for correlation, collinearity, and multicollinearity issues using VIF (Variance Inflation Factor). This step helped us identify variables that were highly

correlated with each other and could potentially distort our model results by inflating standard errors or misrepresenting the effect of predictors. From the correlation matrix and VIF values (as shown from the visualizations in Figures 1 and 2), we found strong correlations between several pairs of variables. For example, *infant.deaths* and *under.five.deaths*, *GDP* and *percentage.expenditure*, and *thinness..1.19.years* and *thinness.5.9.years* were highly correlated with each other (>0.8). To avoid multicollinearity and ensure more stable model estimates, we decided to drop *under.five.deaths*, *infant.deaths*, *thinness..1.19.years*, and *GDP* from our analysis based on their VIF values.

Figures 1 and 2: Correlation matrix and VIF plot used to assess multicollinearity and identify redundant predictors.



Our first model with the raw Life.Expectancy data failed the Breusch-Pagan and Shapiro-Wilk tests. To improve this, we applied a Box-Cox transformation this time around, where we tried a range of lambda values, calculated the log-likelihood for each, and chose the lambda with the highest value. The best lambda was 2, which gave the formula: $(\text{Life.Expectancy}^2 - 1)/2$. This transformation improved the model slightly and made the residuals more balanced, but it did not completely resolve the assumption violations.

Next, we conducted a two-sample t-test to examine whether there was a statistically significant difference in transformed life expectancy between developed and developing countries. The

results showed a significant difference, providing initial evidence that development status has a meaningful impact on transformed life expectancy and helping to support one of our central research questions.

Next, we implemented multiple linear regression models to examine the combined effects of several predictors simultaneously. We used t-tests to assess the significance of individual coefficients and F-tests via ANOVA to evaluate overall model fit. Model performance was quantitatively assessed using R-squared, adjusted R-squared, and root mean squared error (RMSE) metrics and the final model was chosen by weighing those metrics alongside AIC and BIC.

To enhance the accuracy and reliability of our model, we performed diagnostic checks including residual vs. fitted plots, Q-Q plots for normality, and checks for homoscedasticity. We identified multicollinearity as a significant issue in the dataset and explored regularization techniques such as Ridge and LASSO regression to address it. While these methods helped with predictor selection by shrinking less important coefficients, the overall model performance was comparable to that of ordinary least squares (OLS) regression. As a result, we ultimately chose to stick with the simpler OLS model for interpretability and consistency.

Furthermore, we calculated confidence intervals for regression coefficients to assess the precision and robustness of our estimates, and constructed prediction intervals to evaluate the reliability of life expectancy forecasts for new or unseen data, in which the model performs relatively well.

Through this systematic approach, we successfully identified the most influential factors driving transformed life expectancy, including adult mortality rates, immunization coverage, healthcare spending, and years of schooling. The final model (Model 3) provided a balance between interpretability and predictive accuracy, offering meaningful insights into how economic and health-related investments can contribute to improved longevity worldwide.

Data Analysis and Results

After conducting preliminary data analysis and removing variables with high multicollinearity, we developed and tested several multiple linear regression models (along with a polynomial model) to evaluate the impact of various predictor groups on transformed life expectancy (LE_Boxcox) :

- Model 1 (Economic Indicators): LE_Boxcox (Transformed Life Expectancy) ~ Percentage.expenditure + Total.expenditure + Income.composition.of.resource
- Model 2 (Economic and Social Indicators): LE_Boxcox (Transformed Life Expectancy) ~ Percentage.expenditure + Total.expenditure + Income.composition.of.resources + Status + Schooling
- **Model 3 (Combined - Economic, Social, and Health Indicators): LE_Boxcox (Transformed Life Expectancy) ~ Percentage.expenditure + Total.expenditure + Income.composition.of.resources + Status + Schooling + Adult.Mortality + Alcohol + Hepatitis.B + Measles + BMI + Polio + Diphtheria + HIV.AIDS + thinness.5.9.years**
 - Note: *Alcohol* was removed after performing exhaustive subsets and finding that k=13 subset was the best, and not included in the final Model 3.
- Model 4 (Polynomial Combined - (Health Indicators Squared), Economic, Social Indicators): LE_Boxcox (Transformed Life Expectancy) ~ Percentage.expenditure + Total.expenditure + Income.composition.of.resources + Status + Schooling + Adult.Mortality² + Hepatitis.B² + Measles² + BMI² + Polio² + Diphtheria² + HIV.AIDS² + thinness.5.9.years²

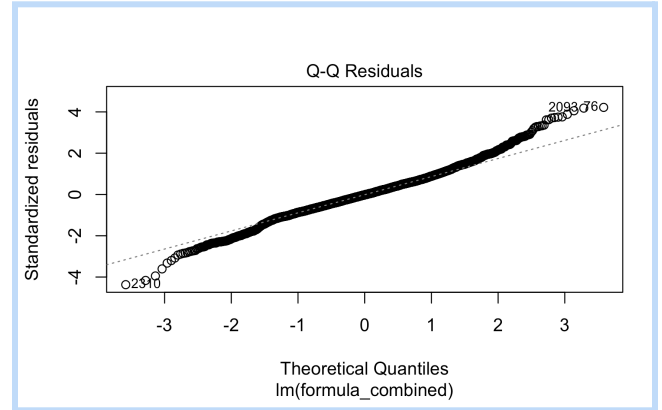
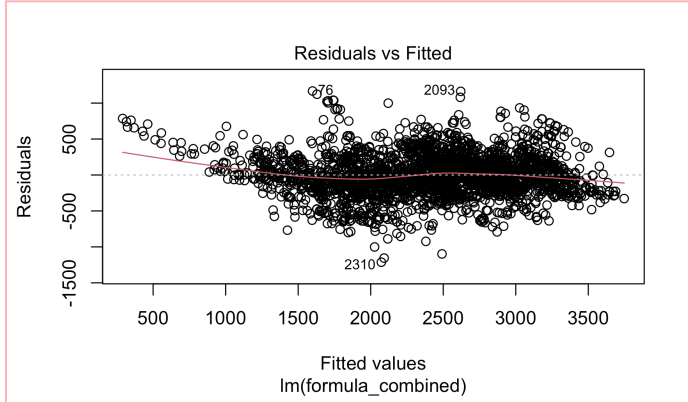
Figure 3: Models' Results - Final Model Comparison

Models	Predictors	Adj. R ²	AIC	BIC	Residual Standard Error	Significance
Economic Indicators	All economic variables	0.526	44038.21	44068.14	434.7	All significant at p < 0.001
Economic + Social	All economic variables + social variables	0.5978	43559.44	43601.34	400.5	All significant at p < 0.001
Combined Model	All Economic + all Social + all Health (without Alcohol) (k= 13 subset on exhaustive subsets output)	0.8057	41429.00	41518.78	278.4	All significant at p < 0.001
Polynomial combined	(All Health variables - measles)squared + (all Economic - total expenditure) + all Social	0.837	40917.85	41049.53	259.9	All except linear and poly Alcohol significant at p < 0.001

We compared these models using Adjusted R², AIC, BIC, and residual standard error. As shown in **Figure 3**, Model 4 (Polynomial Combined) showed the strongest predictive power, with an Adjusted R² of 0.837. However, because Model 4 was polynomial regression and we did not spend much time in class analyzing these, we focused on Model 3 (Combined Model) for further analysis. Model 3, which included economic, social, and health indicators and was created using exhaustive best-subset selection (k = 13), achieved strong performance (Adj. R² = 0.8057) and was easier to interpret within the scope of the course.

After selecting Model 3 as our final model based on performance metrics and interpretability, we proceeded to assess how well it satisfied the assumptions of linear regression. This involved a detailed evaluation of residual patterns, normality, and variance consistency using both visual and statistical diagnostics, as shown in Figures 4 and 5.

Figures 4 and 5: Residual Plots (Model 3)



Model Evaluation

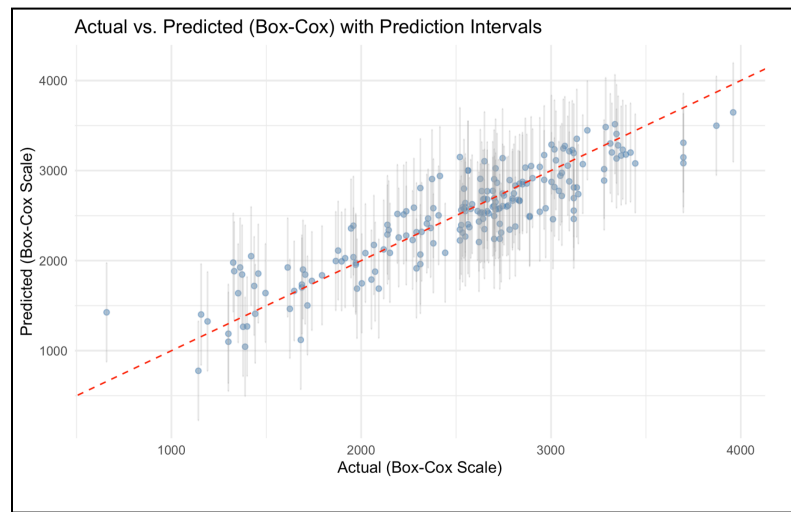
1. **Residual Analysis:** Residual vs. fitted plots indicated slight patterns, with a slight downward curve on the left side and funnel-like shapes at the ends, suggesting heteroscedasticity.
2. **QQ-Plot:** Residuals mostly followed the line, but deviations at the tails hinted at slight non-normality or outliers.
3. **Breuch-Pagan and Shapiro-Wilk's Test:** Formally, the Breusch-Pagan test confirmed heteroscedasticity. The Shapiro-Wilk test showed residuals weren't perfectly normal, likely due to imputation methods or due to the fact that factors affecting life expectancy may be too complex to capture in a linear model.
4. **VIF Analysis:** All predictors' VIF are under 5
5. **Results:** The model shows no serious multicollinearity (all VIF < 5), but two key OLS assumptions are violated (constant error variance and normality of residuals). So the mean predictions are fairly good, yet formal inference (t-tests, confidence intervals, prediction intervals) is less reliable.

Despite minor violations of OLS assumptions observed in the diagnostic tests, we continued our analysis by examining the statistical reliability of the model's estimated coefficients and predictions. Specifically, we computed 95% confidence intervals to assess the precision and directionality of each predictor's effect, as well as prediction intervals to evaluate how accurately

the model can estimate life expectancy for new or unseen countries. These results are summarized in Figures 6 and 7.

Figures 6 and 7: Confidence and Prediction Intervals (Model 3)

	2.5 %	97.5 %
(Intercept)	2537.2536	2596.2236
Adult.Mortality	-179.9982	-153.5951
Hepatitis.B	-35.6595	-12.8930
Measles	-29.9764	-9.1009
BMI	43.3430	69.7338
Polio	33.7057	61.6577
Diphtheria	54.8744	84.1774
HIV.AIDS	-150.3218	-126.4662
thinness.5.9.years	-40.3954	-14.9679
percentage.expenditure	33.8184	57.1406
Total.expenditure	4.0107	25.7573
Income.composition.of.resources	73.6377	108.2998
StatusDeveloping	-185.3286	-118.2066
Schooling	132.2298	168.1306



- Confidence Intervals:** The 95% confidence intervals for Model 3 (all on the Box-Cox scale) show that *Adult.Mortality*, *HIV/AIDS* prevalence, *thinness.5.9.years*, and *Developing* status are strongly negative, whereas *BMI*, immunization rates (*Polio*, *Diphtheria*, *Hepatitis B*, *Measles*), *percentage.expenditure*, *total.expenditure*, and the *income.composition.of.resources* index have clearly positive intervals. However, because the direction of every key predictor remains the same across all three models and the numerical ranges overlap substantially, especially once health variables are added, the confidence intervals are not highly sensitive to model choice; they just tighten as relevant predictors are included, with Model 3 offering the most precise estimate.
- Prediction Intervals:** The 95% prediction intervals illustrate the range in which new, unseen life-expectancy values are likely to fall for a given set of predictors. Because the model draws on both socioeconomic and health indicators (*percentage.expenditure*, *schooling*, immunization coverage), most observed points cluster tightly around the fitted line, showing good overall fit. However, the bands widen towards extremes/ends, where the data are sparse and variability is higher, which means that predictions for very low or high life expectancy countries are less reliable.

Summary & Discussion

This project aimed to identify key economic, social, and health-related factors influencing life expectancy across countries and to develop a predictive regression model using international health and economic data. Our analysis found strong relationships between Box-Cox-transformed life expectancy and key predictors, particularly education and healthcare spending. Model 3 (Combined Model), which included a balanced mix of predictors, performed well with an Adjusted R^2 of 0.8057. Model 4 (Polynomial Combined) demonstrated the strongest overall performance, with the lowest AIC and BIC and the highest Adjusted R^2 (≈ 0.837), indicating more complex relationships between predictors and life expectancy. However, due to its interpretability and alignment with the course emphasis on first-order models, we selected Model 3 for final analysis.

Despite its strengths, Model 3 failed diagnostic tests for normality and constant variance, as indicated by the Shapiro-Wilk and Breusch-Pagan tests. These violations suggest caution when interpreting the model's confidence and prediction intervals, although the model still performed reasonably well across most of the response range. The results also point to the complexity of life expectancy as an outcome that may not be fully captured by a linear model, reinforcing the need to explore more flexible modeling approaches in future analyses.

Overall, our findings highlight that integrating economic, social, and health indicators significantly improves the prediction of life expectancy. These results align with broader public health research linking investments in infrastructure, healthcare systems, and education to improvements in longevity outcomes.

For future work, implementing k-fold cross-validation would provide a more reliable estimate of model performance and help guard against overfitting. Another direction would be to refine the polynomial regression model by experimenting with higher-degree transformations, interaction terms, or regularization techniques to improve both fit and interpretability. Additionally, incorporating external data such as environmental quality metrics, dietary statistics, and health system performance indicators, could help capture important predictors missing from the original dataset. Finally, applying clustering techniques to segment countries by region, income level, or development status may allow for more tailored modeling strategies.

Appendix

1. LOADING DATA

```
#Read the CSV file
life <- read.csv("/Users/charisxiong/Downloads/LifeExpectancy.csv")

#Subset to use later (for additional analyses (like Population
Discrepancy in #3))
life_details <- life %>% dplyr::select(Country, Year, Population)
#Removing Unnecessary Variables (we don't need for main analysis)
data <- subset(life, select = -c(Country, Year))
```

2. EXPLORING MISSING DATA

```
# Calculate missing rows and overall missing data count
missing.rows = dim(data)[1] - dim(na.omit(data))[1]
sprintf("Dataset size: [%s]", toString(dim(data)))

sprintf("Missing rows: %s (%s%%)", missing.rows,
round((missing.rows*100)/dim(data)[1], 2))

# Count missing values for each feature
missing_counts <- data.frame(feature = factor(names(data)),
counts=sapply(data, function(x) sum(is.na(x))))

#Plot missing counts per feature
set_plot_dimensions(16,8)
ggplot(missing_counts,aes(x=reorder(feature, -counts), y=counts,
fill=counts)) + geom_bar(stat="identity") + ggtitle("Missing Counts
in Each Feature") + xlab("Feature") + ylab("Missing count")
+theme(axis.text.x=element_text(angle=25, hjust=1))+theme(text =
element_text(size = 12))+scale_fill_continuous(trans = 'reverse')

#Findings: We have more than 40% of missing data ("Missing rows: 1289
(43.87%)"), so we can apply data imputation (high outliers and
replace with median; low outliers and replace with mean) in section
#5. Most of our missing values are from Population, Hepatitis B, GDP,
Total_expenditure

#Check summary of each variable
variables <- setdiff(names(life), c("Country", "Year"))
#print(summary(data[variables]))
```

3. ADDRESSING POPULATION DISCREPANCY

```
#We have findings that don't make sense (Population has a country
with just 34 people (and more reasons listed below))
```

```

#Discrepancy between different years' population - not possible for
that type of fluctuation (same for Algeria and Thailand, other
countries)
life_details %>%
  filter(Country == "Afghanistan")

#Possible alternative: We can calculate the maximum population per
country. However, it's not possible and doesn't seem logically
correct (Israel, Sri Lanka, Iceland, their max recorded population is
below 100,000, which is way too low compared to actual populations in
2015 (or earlier years))
population_max <- life_details %>%
  group_by(Country) %>%
  summarise(max_population = max(Population, na.rm = TRUE)) %>%
  filter(max_population < 100000)

print(population_max)

#Final Decision: We decided to drop Population because 1) highly
skewed 2) lots of missing values 3) high discrepancies
drop_vars <- c("Population")
data <- data %>% dplyr::select(-all_of(drop_vars))

```

4. CHECKING FOR OUTLIERS FOR EACH FEATURE (VISUALLY)

```

#Strategy: Data imputation, if we have high outliers then replace
with median; low outliers then replace with mean); Findings: Boxplots
that show none/few outliers: Alcohol, BMI,
Income.composition.of.resources

#Plot boxplots for each variable to detect outliers
plot_list <- lapply(names(data), function(var) {
  ggplot(data, aes_string(x = "1", y = var)) +
    geom_boxplot(fill = "lightblue", outlier.colour = "lightblue",
width = 0.3) +labs(title = paste("Boxplot of", var), x = "", y = var)
+theme(axis.text.x = element_blank(), axis.ticks.x =
element_blank(),plot.title = element_text(size = 10),axis.title =
element_text(size = 8),axis.text = element_text(size = 7)))})

# Arrange the plots into pages with 2 rows x 3 columns per page
multi_page_plots <- marrangeGrob(plot_list, nrow = 2, ncol = 3, top =
"Boxplots of Variables")

print(multi_page_plots)

```

5. IMPUTATION

```

# Define variables for imputation:
median_impute <- c(

```

```

  "Life.expectancy", "Adult.Mortality", "Hepatitis.B", "Polio",
  "Diphtheria", "Total.expenditure", "GDP",
  "thinness..1.19.years", "thinness.5.9.years", "HIV.AIDS",
  "percentage.expenditure", "infant.deaths", "Measles",
  "under.five.deaths")
mean_impute <- c("Alcohol", "BMI", "Income.composition.of.resources",
  "Schooling")

# Apply median imputation for variables with high outliers
data[median_impute] <- lapply(data[median_impute], function(x)
  ifelse(is.na(x), median(x, na.rm = TRUE), x))
# Apply mean imputation for variables with low outliers
data[mean_impute] <- lapply(data[mean_impute], function(x)
  ifelse(is.na(x), mean(x, na.rm = TRUE), x))

# Factor conversion for categorical variable
data$Status <- as.factor(data$Status)
# class(data$Status)

```

6. MULTICOLLINEARITY CHECK (CORRELATION, VIF)

```

# Calculate Correlation Matrix
set_plot_dimensions(16, 10)
corr <- round(cor(subset(data, select = -c(Status)), use =
  "complete.obs"), 3)

ggcorrplot(corr, type = "upper", lab = TRUE, outline.color = "black",
  lab_size = 1.5, legend.title = "Correlation") + ggtitle("Correlation
  Matrix") + theme(axis.text.x = element_text(angle = 45, vjust = 1,
  hjust = 1, size = 6), axis.text.y = element_text(size = 6))

# Check multicollinearity using VIF on a linear model
mod <- lm(Life.expectancy ~ ., data = subset(data, select =
  -c(Status)))
vifs <- vif(mod)
print(vifs)

# Plot VIF values with a reference line at VIF = 5
vifs_df <- data.frame(Feature = names(vifs), VIF = vifs)
set_plot_dimensions(16, 8)
ggplot(vifs_df, aes(x = Feature, y = VIF, fill = VIF > 5)) +
  geom_bar(stat = "identity") + scale_y_continuous(trans = "sqrt",
  breaks = c(5, 10, 50, 100)) +
  geom_hline(yintercept = 5, colour = "red", linetype = "dashed",
  size = 1) + ggtitle("VIF per Feature") + xlab("Features") +
  ylab("VIF") + theme_minimal() +
  theme(

```

```

    plot.title = element_text(size = 15, face = "bold"), axis.title.x
= element_text(size = 12, face = "bold"), axis.title.y =
element_text(size = 10, face = "bold"), axis.text.x =
element_text(size = 8, angle = 20, hjust = 1), axis.text.y =
element_text(size = 8), legend.text = element_text(size = 10),
legend.title = element_text(size = 12) ) +scale_fill_brewer(palette =
"Dark2")

```

```

#Drop high VIF variables that were strongly correlated with something
else

```

```

drop_vars <- c("infant.deaths", "thinness..1.19.years", "GDP",
"under.five.deaths")
data <- data %>% dplyr::select(-all_of(drop_vars))

```

```

#Findings: There is a strong correlation [> 0.8] between
infant.deaths and under.fivedeaths; GDP and percentage.expenditure;
thinness.1.19.years and thinness.5.9.years
#We need to avoid multicollinearity issues, so we need to drop one of
each pair. We decided to drop infant.deaths because under.fivedeaths
incorporates infant.deaths plus more (Correlation = 1) and
under.five.deaths because it has such a high VIF that would mess up
our errors
#We decided to drop thinness.1.19 years and GDP because of VIF. As
for income.composition.of.resources and schooling that had relatively
high correlation (0.8), we're going to keep them because VIF doesn't
suggest otherwise

```

```

#Correlation Matrix again
set_plot_dimensions(16, 10)
corr <- round(cor(subset(data, select = -c(Status)), use =
"complete.obs"), 3)
ggcorrplot(corr, type = "upper", lab = TRUE, outline.color = "black",
lab_size = 1.5, legend.title = "Correlation") + ggtitle("Correlation
Matrix") + theme(axis.text.x = element_text(angle = 45, vjust = 1,
hjust = 1, size = 6), axis.text.y = element_text(size = 6))

```

7. Standardize Numeric Predictors

```

num_vars <- setdiff(names(data), c("Status", "Life.expectancy"))
data_scaled <- data
data_scaled[num_vars] <- scale(data[num_vars])

```

8. Box-Cox Analysis

```

# Fit combined model on original y
all_preds <- setdiff(names(data_scaled), "Life.expectancy")

```

```

formula_all <- as.formula(paste("Life.expectancy ~", paste(all_preds,
collapse = " + ")))
mod_all <- lm(formula_all, data = data_scaled)

# Find optimal lambda
bc <- boxcox(mod_all, lambda = seq(-2, 2, by = 0.1))

best_lambda <- bc$x[which.max(bc$y)]
cat("Optimal lambda:", best_lambda, "\n") #2

# Transform y
data_scaled$LE_boxcox <- (data_scaled$Life.expectancy^best_lambda -
1) / best_lambda

```

7. TWO SAMPLE T-TEST

```

#BOXPLOT OF DEVELOPED AND DEVELOPING COUNTRIES' LIFE EXPECTANCIES OF
TRANSFORMED Y
set_plot_dimensions(10, 6)
ggplot(data_scaled, aes(x = Status, y = LE_boxcox, fill = Status)) +
  geom_boxplot() +
  labs(title = "Transformed Life Expectancy by Country Status", x =
"Status", y = "Transformed Life expectancy")

#TWO SAMPLE T-TEST (DEVELOPED AND DEVELOPING)
#H0: The average life expectancy is the same for developing and
developed countries; Ha: The average life expectancy differs.
#Findings: Since p-value < 0.05, 95% confidence interval does not
include 0, we can conclude that, on average, developed countries are
significantly different from developing countries in terms of life
expectancies (transformed)

t_test_result <- t.test(LE_boxcox ~ Status, data = data_scaled)
print(t_test_result)

```

8. BUILDING MULTIPLE LINEAR REGRESSION MODELS

```

# Define variable groups for modeling
econ_vars <- c("percentage.expenditure", "Total.expenditure",
"Income.composition.of.resources")
social_vars <- c("Status", "Schooling")
health_vars <- c("Adult.Mortality", "Alcohol", "Hepatitis.B",
"Measles", "BMI", "Polio", "Diphtheria", "HIV.AIDS",
"thinness.5.9.years")

# Model 1: Transformed Life Expectancy ~ Economic Indicators
# Hypothesis: Better economic conditions (such as
Income.composition.of.resources,

```

```

#percentage.expenditure) are associated with higher transformed life
expectancy.
formula_econ <- as.formula(paste("LE_boxcox ~", paste(econ_vars,
collapse = " + ")))
model1 <- lm(formula_econ, data = data_scaled)
summary(model1)

# Model 2: Transformed Life Expectancy ~ Economic Indicators + Social
Indicators
# Hypothesis: Better economic conditions and higher schooling +
#status as a developed/developing country are associated with a
higher transformed life expectancy
econ_social_vars <- c(econ_vars, social_vars)
formula_social_econ <- as.formula(paste("LE_boxcox ~",
paste(econ_social_vars, collapse = " + ")))
model2 <- lm(formula_social_econ, data = data_scaled)
summary(model2)

# Model 3: Combined Model: LE_boxcox ~ Health + Economic + Social
Indicators
# Hypothesis: A model integrating health, economic, and social
variables provides the best overall prediction.
all_vars <- c(health_vars, econ_vars, social_vars)
formula_combined <- as.formula(paste("LE_boxcox ~", paste(all_vars,
collapse = " + ")))
model3 <- lm(formula_combined, data = data_scaled)
summary(model3)

# Model 4: POLYNOMIAL Combined Model: LE_boxcox ~ Health^2 + Economic
+ Social Indicators

# Hypothesis: A model integrating health, economic, and social
variables, while incorporating polynomial terms for the
health-related

# variables provides the best overall prediction.

health_vars_poly <- c("Adult.Mortality", "Alcohol", "Hepatitis.B",
"BMI", "Polio", "Diphtheria", "HIV.AIDS", "thinness.5.9.years")

health_poly_vars <- sapply(health_vars_poly, function(x)
paste("poly(", x, ", 2)", sep=""))

all_vars_poly <- c(health_poly_vars, "percentage.expenditure",
"Income.composition.of.resources", social_vars)

formula_poly <- as.formula(paste("LE_boxcox ~", paste(all_vars_poly,
collapse = " + ")))

```

```

model_poly <- lm(formula_poly, data = data_scaled)
summary(model_poly)

# SHOWING ALL OF THE SUBSETS:

library(leaps)

all_vars <- c(health_vars, econ_vars, social_vars)

formula_combined <- as.formula(paste("LE_boxcox ~", paste(all_vars,
collapse = " + ")))

allmodels <- regsubsets(formula_combined, data = data_scaled, nbest =
1, nvmax = length(all_vars))

summary(allmodels)

#this section creates the dataframe of all the AIC, BIC, adjusted R
squared values

all_vars <- c(health_vars, econ_vars, social_vars)

form_combined <- as.formula(
  paste("LE_boxcox ~", paste(all_vars, collapse = " + "))
)

allmodels <- regsubsets(form_combined, data = data_scaled,
                        nbest = 1, nvmax = length(all_vars))

reg.sum <- summary(allmodels) # rebuild for this search

fix_factor <- function(v) sub("^Status.*", "Status", v)

nv <- length(reg.sum$rss)

ic_list <- lapply(1:nv, function(k){
  vars <- names(coef(allmodels, id = k))[-1] # predictors in
best k-subset
  vars <- unique(fix_factor(vars))
  #fml <- reformulate(vars, response = "LE_boxcox")

```

```

    fml <- as.formula(paste("LE_boxcox ~", paste(vars, collapse = " +
")))

    fit <- lm(fml, data = data_scaled)

    c(k= k,AIC = AIC(fit),BIC = BIC(fit),AdjR2
=summary(fit)$adj.r.squared)

})

```

```

ic_table <- as.data.frame(do.call(rbind, ic_list))

ic_table$AIC <- round(ic_table$AIC, 1)
ic_table$BIC <- round(ic_table$BIC, 1)
ic_table$AdjR2 <- round(ic_table$AdjR2, 4)

```

```

print(ic_table, row.names = FALSE)

```

#end of section which creates the dataframe of all the AIC, BIC, adjusted R squared values

#Findings:

- Model 1 included economic indicators like percentage health expenditure, total expenditure, and income composition of resources. All variables were highly significant, and the R^2 is 0.5267 and adjusted R-squared was about 0.5263, which suggests that economic factors explain about 53% of variation in transformed life expectancy.

- Model 2 focused on combining both economic and social indicators to see if they were significant in determining transformed life expectancy.

#The variables in here were highly significant, with the R^2 being 0.5984 and the adjusted R-squared was about 0.5978, which indicates that adding social factors improved explanatory power by approximately 7%.

- Model 3 combined health, economic, and social variables together. This model produced the highest R^2 (0.8066) adjusted R-squared of about 0.8057,

#showing that combining health, economic, and social factors provides the strongest predictive power for transformed life expectancy. However, not all variables in this combined model were

statistically significant. For example, alcohol consumption had p-values greater than 0.05.

#

- Model 4 polynomial function which combined health, economic, and social variables together. All of the variables in the "health" category were squared to account for non-linear relationships between these health indicators and transformed life expectancy. This model shows that non-linear relationships in health variables should be considered for a more accurate prediction of life expectancy, and should be considered alongside economic and social variables

9. BACKWARD STEPWISE SELECTION

```
#Drop Life.expectancy
data_scaled <- data_scaled %>%dplyr::select(-Life.expectancy)

step_model3 <- stepAIC(model3, direction = "backward", trace = FALSE)
summary(step_model3)

#Findings: Only variable we dropped was Alcohol
```

10. DROPPING INSIGNIFICANT VARIABLES FROM MODEL 3 BASED ON BACKWARD STEPWISE SELECTION

```
#Dropping non significant variables: Alcohol
all_vars <- setdiff(all_vars, c("Alcohol", "Life.expectancy"))

formula_combined <- as.formula(paste("LE_boxcox ~", paste(all_vars, collapse = " + ")))
model3_refined <- lm(formula_combined, data = data_scaled)
summary(model3_refined)

#All variables are statistically significant at the 0.1%, 1%, and 5% levels
#Findings: R^2: 0.8066, Adjusted R-squared: 0.8057
```

11. ACTUAL VERSUS PREDICTED

```
#Get predictions on the Box-Cox scale
pred1 <- predict(model1, newdata = data_scaled)
pred2 <- predict(model2, newdata = data_scaled)
pred3 <- predict(model3_refined, newdata = data_scaled)
pred4 <- predict(model_poly, newdata = data_scaled)

#Build wide dataframe (Box-Cox)
df_all <- data.frame(actual_bc = data_scaled$LE_boxcox, Model1_bc = pred1, Model2_bc = pred2, Model3_bc = pred3, Model4_bc = pred4)
```

```

#Melt (Box-Cox)
df_long_bc <- melt(df_all,id.vars = "actual_bc",measure.vars =
c("Model1_bc","Model2_bc","Model3_bc", "Model4_bc"),variable.name =
"Model", value.name= "predicted_bc")

#Plot (Box-Cox)
ggplot(df_long_bc, aes(x = actual_bc, y = predicted_bc)) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed") +
  geom_point(alpha = 0.5) +
  facet_wrap(~ Model, ncol = 3) +
  labs(x = "Actual (Box-Cox Scale)",
       y = "Predicted (Box-Cox Scale)",
       title = "Actual vs Predicted by Model (Box-Cox Scale)") +
  theme_minimal()

# Predictions with intervals on Box-Cox scale
predictions <- predict(model3_refined, newdata = data_scaled,
interval = "prediction")
df_pred_interval <- data.frame(actual_bc = data_scaled$LE_boxcox,
                              predicted_bc = predictions[, "fit"],
                              lower_bc = predictions[, "lwr"],
                              upper_bc = predictions[, "upr"])

# Plot Box-Cox prediction intervals
subset_bc <- df_pred_interval[sample(nrow(df_pred_interval), 200), ]

ggplot(subset_bc, aes(x = actual_bc, y = predicted_bc)) +
  geom_point(alpha = 0.5, color = "steelblue") +
  geom_errorbar(aes(ymin = lower_bc, ymax = upper_bc), alpha = 0.3,
color = "darkgrey") +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color =
"red") +
  labs(title = "Actual vs. Predicted (Box-Cox) with Prediction
Intervals",x = "Actual (Box-Cox Scale)", y = "Predicted (Box-Cox
Scale)") + theme_minimal()

```

12. ANOVA + CONFIDENCE INTERVALS

```

#ANOVA (Compare model fit)
anova_tbl <- anova(model3_refined)
# round every numeric column to 4 places
anova_tbl[] <- lapply(anova_tbl, function(col)
  if (is.numeric(col)) round(col, 4) else col

```

```
)
print(anova_tbl)

#Confidence Intervals for coefficients
ci <- confint(model3_refined)
ci_4dec <- round(ci, 4)
print(ci_4dec)

#Findings:
#ANOVA confirms that almost all variables in our final model are
statistically significant contributors to explaining transformed life
expectancy. Confidence intervals provide a range of plausible values
for each coefficient. They all do not contain 0, which further
indicates that they are statistically significant
```

```
13. COMPARING MODELS USING AIC, BIC, ADJUSTED R_SQUARED
models_list <- list(model1, model2, model3, model_poly)

model_names <- c("Economic Indicators", "Economic and Social
Indicators", "Combined Model", "Combined Model - Polynomial")

aic_values <- sapply(models_list, AIC)
bic_values <- sapply(models_list, BIC)

model_comparison <- data.frame(Model = model_names, AIC = aic_values,
BIC = bic_values, Adj_R_squared = sapply(models_list, function(m)
summary(m)$adj.r.squared))

model_comparison$Adj_R_squared <-
round(model_comparison$Adj_R_squared , 4)
print(model_comparison)
```

```
14. REGULARIZATION TECHNIQUES (RIDGE, LASSO)
# Preparing data for regularization
x <- model.matrix(LE_boxcox ~ ., data = data_scaled)[, -1]
y <- data_scaled$LE_boxcox

#Ridge
ridge_cv <- cv.glmnet(x, y, alpha = 0, nfolds = 10)
ridge_best_lambda <- ridge_cv$lambda.min
ridge_mod <- glmnet(x, y, alpha = 0, lambda = ridge_best_lambda)

#LASSO
lasso_cv <- cv.glmnet(x, y, alpha = 1, nfolds = 10)
lasso_best_lambda <- lasso_cv$lambda.min
lasso_mod <- glmnet(x, y, alpha = 1, lambda = lasso_best_lambda)
```

```

# Plot cross-validation curves
par(mfrow = c(1,2))
plot(ridge_cv, main = "Ridge CV")
plot(lasso_cv, main = "LASSO CV")

# Coefficient comparison
coef_ols <- as.matrix(coef(model3_refined))
coef_ridge <- as.matrix(coef(ridge_mod))
coef_lasso <- as.matrix(coef(lasso_mod))

all_vars <- unique(c(rownames(coef_ols), rownames(coef_ridge)))

get_coef <- function(mat, v) {
  if (v %in% rownames(mat)) mat[v,1] else 0
}

coef_comp <- data.frame(Variable = all_vars, OLS= sapply(all_vars,
function(v) get_coef(coef_ols,v)), Ridge = sapply(all_vars,
function(v) get_coef(coef_ridge, v)), LASSO = sapply(all_vars,
function(v) get_coef(coef_lasso, v)))
print(coef_comp)

perf_df <- data.frame(
  Model = c("OLS", "Ridge", "LASSO"),
  RMSE = c(rmse(y, pred_ols), rmse(y, pred_ridge), rmse(y,
pred_lasso)),
  R_squared = c(1 - sum((y - pred_ols)^2) / sum((y - mean(y))^2), 1 -
sum((y - pred_ridge)^2) / sum((y - mean(y))^2), 1 - sum((y
-pred_lasso)^2) / sum((y - mean(y))^2)))

# Number of observations
n <- nrow(data_scaled)

# Number of predictors (excluding intercept) in each model
p_ols <- length(coef(model3_refined)) - 1
p_ridge <- length(coef(ridge_mod)) - 1
p_lasso <- sum(coef(lasso_mod) != 0) - 1

# Compute Adjusted_R^2 for each
perf_df$Adj_R_squared <- c(1 - (1 -
perf_df$R_squared[perf_df$Model=="OLS"]) * (n - 1) / (n - p_ols-
1), 1 - (1 - perf_df$R_squared[perf_df$Model=="Ridge"]) * (n - 1) / (n
- p_ridge - 1), 1 - (1 - perf_df$R_squared[perf_df$Model=="LASSO"]) *

```

```
(n - 1) / (n - p_lasso - 1)
)
perf_df$R_squared <- round(perf_df$R_squared, 4)
perf_df$Adj_R_squared <- round(perf_df$Adj_R_squared, 4)
```

```
print(perf_df)
```

#Findings: All three approaches—ordinary least squares, Ridge, and LASSO—now perform virtually identically (RMSE is around 277.7, R^2 is around 0.806). The fact that neither Ridge nor LASSO improves on OLS indicates there's little residual multicollinearity or overfitting to correct. In this case, the simplest solution (plain OLS) is just as predictive as any regularized alternative, so it's reasonable to stick with OLS for both clarity and interpretability.

15. RESIDUAL CHECKS & MODEL DIAGNOSTICS

```
#Findings:
```

```
#Residual vs Fitted - Our model appears to fit the bulk of the data
really well, but there is a slight non-linearity and (funnel shape?)
at higher fitted values and outliers (suggests mild
heteroscedasticity) these are natural given the international life
expectancy data
```

```
#QQ Plot - we expect that the points are roughly following along the
straight line, which seems correct in this case (except for
deviations at the ends, which is common), which confirms the
residuals are roughly normally distributed with skewness/heavy tails
at least visually, but we can run some tests to confirm so
```

```
plot(model3_refined, which = 1)
```

```
plot(model3_refined, which = 2)
```

```
# Breusch-Pagan Test (Check for heteroscedasticity) - Strong evidence
of heteroscedasticity, which means that standard errors could be
wrong -> p-values could be misleading (Model can predict well, but
inference (like p-values & CI's) may not be valid unless fixed;
bptest(model3_refined)
```

```
# Shapiro-Wilk Test (Test for normality of residuals) - Residuals not
perfectly normal. Due to large sample size, this is common and not
critical; With large n, Central Limit Theorem helps, so not a huge
problem
shapiro.test(residuals(model3_refined))
```

```
#VIF (multicollinearity) - All < 5; so there is no concern here
vif(model3_refined)
```

16. FINAL MODEL PERFORMANCE METRICS

```
#Model Performance Metrics (RMSE & R-squared)
rmse(actual = data_scaled$LE_boxcox, predicted =
predict(model3_refined))

## [1] 277.6886

# R-squared - 0.8066
# Findings: Approximately 80.66% of the variation in transformed life
expectancy can be explained by the selected predictors in the model.
The RMSE is 277.6886 indicating that on average, the model's
predictions deviate from the actual transformed life expectancy
values by about 277.6886 units.
#These results reflect a relatively strong model fit given the
complexity of global health data
```

References

- Kutner, M.H., Nachtsheim, C.J., & Neter, J. (2004). Applied Linear Regression Models (4th ed.). McGraw-Hill.
 - Used for: Referenced material for guidelines on model building, diagnostic techniques, handling multicollinearity, Box-Cox transformations
- Kumar, R. (2015). Life Expectancy (WHO). Kaggle.
<https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who>
 - Used for: the primary dataset: variable definitions
- World Health Organization. (2023). Life Expectancy and Healthy Life Expectancy.
<https://www.who.int/data/gho>
 - Used for: clarifying how WHO calculates life-expectancy indicators, and providing context for discussion of health-care quality gaps.
- Roser, M., Ritchie, H., & Ortiz-Ospina, E. (2023). Life Expectancy. Our World in Data.
<https://ourworldindata.org/life-expectancy>
 - Used for: historical life-expectancy trends and visual examples
- Centers for Disease Control and Prevention (CDC). (2023). Life Expectancy in the U.S. Dropped Again. <https://www.cdc.gov/nchs/pressroom>
 - Used for: Supplied recent U.S. figures (infant mortality, insurance coverage)

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Self-reflections

Charis Xiong - I spent approximately 15 hours on this project, focusing on identifying unreliable predictors, handling missing data, and preparing the dataset for modeling. This deepened my understanding of multiple linear regression, particularly in checking key assumptions, addressing multicollinearity, and using Box-Cox transformations to improve model fit. One challenge we encountered was improving model assumptions. Despite using tools like VIF to manage multicollinearity and transformations, we still failed certain assumptions and tests, which led to critical reflection on why these assumptions weren't met. One issue that went wrong was our attempt to back-transform predictions for interpretability. Initially, we considered presenting results on the original life expectancy scale, but mixing scales created confusion, and the nonlinear Box-Cox transformation distorted confidence and prediction intervals. As a result, we decided against this approach. If I were to redo the project, I would spend more time experimenting with different transformation strategies, such as Box-Cox variants, log or power transforms, to better balance model assumptions and interpretability. My advice for future students is to dedicate time to preliminary work, especially when dealing with imperfect datasets. Apply class concepts as you progress, and starting early will ensure a smoother process and build confidence in your analysis.

Meaghan Ramlakhan - During this project, I contributed around 15-16 hours researching and selecting the dataset, drafting the proposal, designing the presentation slides, and conducting model diagnostics. I focused on evaluating key assumptions like normality, constant variance, and multicollinearity, using tools like the Shapiro-Wilk test, the Breusch-Pagan test, VIF scores, and residual plots. This process helped me gain a deeper understanding of how assumption violations impact the reliability of statistical models and encouraged a more critical and analytical approach to model evaluation. I also helped refine the visual design of our slides to ensure that our technical results were communicated clearly and effectively. The proposal stage was particularly helpful in clarifying our research direction and providing a solid foundation for the modeling phase. If I had more time, I would explore interactive visualizations to better understand variable relationships and incorporate domain knowledge earlier to guide model selection. My advice to future students is to start early, be thorough with diagnostic testing, and remember that understanding the data is as crucial as running the analysis itself.

Pinar Targil - Throughout this project, I devoted approximately 15 hours to collaborative data analysis, model selection, and refining our approach. My main focus was on dataset exploration and making decisions regarding variable imputation and exclusion, especially for columns with extreme skew or inconsistencies. I also selected the optimal regression model using exhaustive best-subset selection, comparing predictor combinations based on adjusted R^2 , AIC, BIC, and model simplicity. I developed a Polynomial model and analyzed its R^2 , AIC, and BIC, which I included in the model comparisons. If I had more time, I would explore additional validation techniques like k-fold cross-validation and experiment with interaction terms. My advice to future students is to begin early and be prepared to refine your models iteratively. Success in projects like this comes from both technical precision and collaborative problem-solving.